Adil Osman

Data Engineer

aosman10020@gmail.com • +44 7538 046 026 • London, UK • Right to Work
github.com/aosman101 • linkedin.com/in/adil-osman-303185297

PROFESSIONAL SUMMARY

Data Engineer with nearly three years of experience building production-grade ETL/ELT pipelines, real-time streaming and CDC systems, and analytics models with Python, SQL, dbt, Apache Airflow, and BigQuery. Most recently delivered 10+ end-to-end data platforms as a freelance consultant, cutting cloud warehouse query costs by ~40% and raising automated data quality test coverage to 95%+. Earlier roles delivered 99% SLA adherence, an 8x improvement in data freshness, and a 38% reduction in analysis cycle time. Hands-on with Kafka-compatible streaming (Redpanda), CDC (Debezium), and OLAP (ClickHouse), with exposure to Databricks/PySpark, Delta Lake, and AWS. Known for observable, well-documented data systems that technical and non-technical stakeholders can rely on.

WORK EXPERIENCE

Data Engineer & Analytics Engineer | Freelance (Self-Employed) · Remote **Jan 2026 – Sep 2026**

- Delivered 10+ end-to-end data platforms as an independent consultant, turning fragmented multi-source client data into reliable, analytics-ready pipelines with measurable SLAs and full lineage.
- Reduced cloud warehouse query costs by ~40% by re-engineering transformations into incremental dbt models with partitioning on BigQuery.
- Increased automated data quality test coverage to 95%+ with dbt tests and Great Expectations, catching schema and freshness issues before they reached stakeholders.
- Cut data preparation defects by ~30% by enforcing deduplication, schema contracts, and validation at ingestion.
- Engineered near real-time CDC analytics (Postgres → Debezium → Redpanda/Kafka → ClickHouse) and Airflow lakehouses on partitioned Parquet (dbt + DuckDB), with OpenLineage lineage and GitHub Actions CI/CD.

Data Engineer | Somalis in Tech **Apr 2024 – Jul 2025**

- Developed Python ingestion scripts and scheduled ETL jobs, improving data freshness from ~6 hours to ~45 minutes and automating ~65% of manual reporting.
- Built and optimised SQL transformations to produce analytics-ready tables for recurring KPI reporting, with consistent naming conventions and clear definitions.
- Optimised pipeline logic to meet weekly reporting SLAs, achieving 99% on-time delivery with P95 end-to-end job latency under 60 minutes.
- Introduced monitoring and triage routines (run logging, documented checks, escalation steps) to reduce time-to-diagnosis for failed jobs and speed up recovery.
- Partnered with programme leads to define engagement KPIs and delivered Tableau/ThoughtSpot dashboards, supporting a 22% uplift in attendance for priority programmes.
- Mentored peers on data engineering best practices, SQL patterns, and pipeline design, fostering a culture of documentation and reproducibility within the team.

Data Analyst | HQ Analytics **Jun 2023 – Feb 2024**

- Engineered repeatable data preparation workflows in Python and SQL to clean, validate, and transform large datasets, reducing analysis cycle time for recurring workflows by 38%.
- Automated data extraction pipelines with scheduled jobs and reusable scripts, saving ~6 hours of manual effort per week and improving consistency.
- Collaborated on SQL data models for analytics-ready tables, improving typical query performance by ~48% through partitioning and clustering strategies.
- Performed data profiling and source-system analysis (null checks, duplicates, distribution checks) and documented assumptions to support correct interpretation downstream.
- Worked with analysts to clarify reporting requirements and ensure warehouse schemas supported dashboards and ad-hoc analysis.

TECHNICAL SKILLS

Data Engineering: ETL/ELT pipelines, API/file ingestion, incremental loading, CDC (Debezium), Kafka-compatible event streaming (Redpanda), dimensional modelling, and performance tuning. Proficient in orchestration with Apache Airflow, dbt transformations, and SQL optimisation.

Platforms & Tooling: BigQuery, ClickHouse, DuckDB, PostgreSQL, Snowflake (exposure), Databricks/PySpark (exposure). AWS (S3, IAM, EC2, RDS). Terraform (exposure). Git/GitHub with CI/CD via GitHub Actions.

Quality, Observability & BI: dbt tests and Great Expectations for data quality; data lineage with OpenLineage/Marquez. Dashboards in Tableau, Power BI, and ThoughtSpot; visualisations with Plotly and Matplotlib.

EDUCATION

BSc Data Science & Computing — First Class Honours **Graduated December 2025**

Birkbeck, University of London

- **Key Modules:** Database Management (90%), Artificial Intelligence & Machine Learning (84%), Software & Programming II (84%), Data Structures & Algorithms (84%).
- **Programming Languages:** Python, SQL, R, Java, JavaScript, HTML/CSS, PowerShell, and Linux/Bash.

PROJECTS

StreamShop CDC Analytics Stack (Postgres → Debezium → Redpanda/Kafka → ClickHouse)

Built an end-to-end, production-style CDC pipeline where Postgres changes stream through Debezium into a Kafka-compatible broker (Redpanda) and land in ClickHouse via a Python sink. Modelled analytics marts and an SCD Type 2 snapshot in dbt, with schema registry and CI to validate models and contracts.

Tech Stack: Postgres, Debezium, Redpanda (Kafka API), Kafka, ClickHouse, dbt, Python, Docker Compose, GitHub Actions

London Air Quality & Weather Lakehouse

Built a containerised local lakehouse retrieving hourly OpenAQ and Open-Meteo data via Airflow, landing raw JSON in MinIO (S3-compatible), upserting flattened rows into Postgres, modelling staging-to-marts in dbt with incremental patterns, validating with Great Expectations, and serving Metabase dashboards.

Tech Stack: Airflow, Python, MinIO (S3), Postgres, dbt, Great Expectations, Metabase, Docker

TfL Real-Time Lakehouse

Built a real-time lakehouse on live TfL arrivals using Airflow (Celery) and Marquez. Ingestion writes date-partitioned Parquet snapshots, dbt and DuckDB build analytics marts, Great Expectations runs inline validations, and OpenLineage metadata is emitted for end-to-end lineage visibility.

Tech Stack: Airflow, Python, Parquet, DuckDB, dbt, Great Expectations, OpenLineage/Marquez, Docker

Premier League Analytics Warehouse (BigQuery + dbt + CI)

Created an end-to-end analytics warehouse on Google BigQuery. Python extracts openfootball JSON into NDJSON, loads raw data into BigQuery, and dbt builds cleaned staging models and dimensional-style marts (teams, match results, season stats) with schema tests and CI-friendly configuration.

Tech Stack: Python, BigQuery, dbt Core, GitHub Actions, SQL, NDJSON